

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)		Reasonable attempt at sinusoidal curve (1) Accurate curve drawn with maximum and minimum at correct points (1) N.B. Look out for points being missed, trough in the right place, starting too low		2		2	1	2
	(b)		Period – the time taken for one complete cycle of the oscillations (1) 0.9 ± 0.01 [s] (1)	1	1		2		2
	(c)		$y = 1.5 \cos\left(\frac{2\pi}{0.9[1]}t \pm \pi(1)\right)$ cm ecf on period Don't accept 180		2		2	1	1
	(d)	(i)	Good tangent (1) Substitution of values for the gradient ecf (1) Max velocity i.e. 10.5 ± 1.0 [cm s ⁻¹] (1) ecf	1 1	1		3	2	3
		(ii)	Max velocity of 10 [cm s ⁻¹] ecf (1) [Sinusoidal curve with] appropriate period ecf (1) Appropriate phase for starting point i.e. sin curve (1)		3		3	2	3
	(e)	(i)	Acceleration is [directly] proportional to displacement (1) And is directed towards a fixed point (1)	2			2		
		(ii)	Substitution: $T = 2\pi\sqrt{\frac{m}{k}}$ i.e. $0.9 = 2\pi\sqrt{\frac{0.030}{k}}$ (1) ecf Rearrangement: $k = \frac{4\pi^2 \times 0.030}{0.81}$ (1) Answer = 1.46 N m ⁻¹ **unit mark** (1)	1	1 1		3	2	3

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	(f)		Attempt at $\frac{1}{2}mv^2$ i.e. $0.5\text{ m } v_{\text{max}}^2 = 0.5 (0.030) (0.1)^2(1)$ $= 1.5 \times 10^{-4} \text{ J}(1)$ Extension at equilibrium $(= \frac{mg}{k} = \frac{(0.030)(9.81)}{1.46} = 0.20 \text{ m } (1)$ Adding 1.5 cm [= 0.215 m] (1) Attempt at $\frac{1}{2}kx^2$ i.e. $= 0.5 k x^2 0.5 (1.46)(0.215)^2[1 \text{ ecf}]$ $= 0.034 \text{ [J]}]$ so therefore claim is correct (1)			6	6	6	6
			Question 4 total	6	11	6	23	14	20